

Module 07: Communication — Coupled Inference, Shared Generative Models, and Multi-Agent Free Energy

Learning Objectives

1. Formalize communication as **coupled inference**: two agents mutually predicting each other's behavior.
2. Derive the **multi-agent free energy** and show how it decomposes into individual and interaction terms.
3. Develop the mathematics of **generalized synchrony** between coupled dynamical systems.
4. Connect multi-agent Active Inference to **information-theoretic** measures of communication.

Introduction

When two Active Inference agents interact, each treats the other as part of its environment — a generative model that must be inferred. Communication arises when both agents are simultaneously inferring each other's hidden states, producing a coupled dynamical system. This module develops the mathematics of multi-agent Active Inference.

Key Concepts

1. Coupled Inference Between Two Agents

Consider agents A and B. Agent A's generative model includes a model of agent B:

$$p_A(o_A, s_A, s_B) = p_A(o_A | s_A, s_B) \cdot p_A(s_A) \cdot p_A(s_B)$$

Agent A treats B's states s_B as hidden causes of its own observations o_A . Symmetrically, agent B models agent A. Each agent minimizes its own free energy:

$$F_A = E_{\{q_A\}}[\ln q_A(s_A, s_B) - \ln p_A(o_A, s_A, s_B)] \quad F_B = E_{\{q_B\}}[\ln q_B(s_A, s_B) - \ln p_B(o_B, s_A, s_B)]$$

The coupling arises because A's actions influence B's observations and vice versa, creating a feedback loop.

2. Multi-Agent (Joint) Free Energy

For a system of N agents, the **joint free energy** is:

$$F_{\text{joint}} = \sum_i F_i - I_{\text{mutual}}$$

where F_i is agent i 's individual free energy and I_{mutual} captures the mutual information between agents' states. Joint free energy minimization implies:

1. Each agent minimizes its own free energy (individual inference)
2. The agents' beliefs become mutually consistent (generalized synchrony)

The interaction term measures the degree to which the agents' generative models align — successful communication corresponds to low interaction free energy (high mutual predictability).

3. Generalized Synchrony

Two dynamical systems x_1 and x_2 are in **generalized synchrony** if there exists a smooth function h such that:

$$x_2(t) = h(x_1(t)) \text{ for all } t$$

For coupled Active Inference agents, generalized synchrony means the agents' internal states (sufficient statistics) become functionally related — each agent's beliefs can be predicted from the other's. This is the mathematical formalization of "mutual understanding."

The synchronization manifold is stabilized when the coupling strength exceeds the Lyapunov exponent of the uncoupled systems — a condition analogous to "paying enough attention to each other."

4. Information-Theoretic Communication Measures

Communication between Active Inference agents can be quantified using information-theoretic measures:

- **Mutual information:** $I(s_A; s_B | o)$ measures how much knowing one agent's states tells us about the other's states, given the shared observations. Successful communication maximizes this quantity.
- **Transfer entropy:** $TE(A \rightarrow B) = H(s_{B_future} | s_{B_past}) - H(s_{B_future} | s_{B_past}, s_{A_past})$ measures the directed flow of information from A to B — how much A's past states reduce uncertainty about B's future states.
- **Channel capacity:** The maximum rate at which agents can communicate is bounded by the mutual information between their action and observation channels: $C = \max I(a_A; o_B)$, where the maximum is over the distribution of A's actions.

5. Worked Example: Two Coupled Gaussian Agents

Consider two agents with scalar states and linear coupling:

Agent A: $ds_A/dt = -\kappa_A \cdot s_A + c_{AB} \cdot s_B + \xi_A$ (where $\xi_A \sim N(0, \sigma_A^2)$) **Agent B:** $ds_B/dt = -\kappa_B \cdot s_B + c_{BA} \cdot s_A + \xi_B$ (where $\xi_B \sim N(0, \sigma_B^2)$)

The joint free energy under mean-field approximation:

$$F_{\text{joint}} = \frac{1}{2}\pi_A \cdot (o_A - \mu_A)^2 + \frac{1}{2}\pi_B \cdot (o_B - \mu_B)^2 + \frac{1}{2}\lambda_A \cdot (\mu_A - c_{AB} \cdot \mu_B)^2 + \frac{1}{2}\lambda_B \cdot (\mu_B - c_{BA} \cdot \mu_A)^2 + \dots$$

At steady state, the belief updates satisfy:

$$\mu_A^* = (\pi_A \cdot o_A + \lambda_A \cdot c_{AB} \cdot \mu_B) / (\pi_A + \lambda_A) \quad \mu_B^* = (\pi_B \cdot o_B + \lambda_B \cdot c_{BA} \cdot \mu_A) / (\pi_B + \lambda_B)$$

Each agent's belief is a precision-weighted combination of its own sensory evidence and its prediction of the other agent. The coupling precisions λ_A and λ_B control "how much each agent listens to the other."

6. Scaling to Multi-Agent Systems

For $N > 2$ agents, the mathematics generalizes through **mean field factorization**:

$$q(s_1, \dots, s_N) \approx \prod_i q_i(s_i)$$

Each agent infers only its own states and a summary of the population. This yields tractable inference even in large groups and connects to **collective behavior**: flocking, herding, and social norms emerge as attracting solutions of coupled free energy minimization (Friston & Frith, 2015).

Derivation Exercises

1. For two coupled Gaussian agents, derive the joint free energy F_{joint} and show how it separates into individual and interaction terms.
2. Derive the condition for generalized synchrony: when does the synchronization manifold become an attractor?
3. Show that communication minimizes joint free energy by aligning the agents' generative models.
4. Compute the transfer entropy $TE(A \rightarrow B)$ for the two coupled linear-Gaussian agents and show it depends on the coupling strength c_{BA} .

Conclusion

Communication is coupled inference — two agents minimizing their joint free energy through reciprocal prediction. The mathematics of generalized synchrony and information theory provides the formal framework for understanding how mutual understanding emerges from the dynamics of interacting generative models. Module 08 completes the mathematical framework with planning and sophisticated inference.

Further Reading

- Friston, K. & Frith, C. (2015). A duet for one. *Consciousness and Cognition*, 36, 390-405.

- Schreiber, T. (2000). Measuring information transfer. *Physical Review Letters*, 85(2), 461.
- Vasil, J. et al. (2020). A world unto itself: Human communication as active inference. *Frontiers in Psychology*, 11, 417.